

Verification and Construction as Coupled Ladders

The P versus NP Verdict Is Not Frame-Internal

The circumscription obstruction in its purest form: both verdicts tail-shaped, the barriers as the domain's own frame theorems, the self-referential seam, the corrected reading of the Λ -axiom argument, and the priced specification

Lu Semita · EmergenceByDesign Drawing on the author's P vs NP corpus (the Λ -NSAF musings, Parts 1–2; the DST fine-grained traversal and its Localization Theorem), the Semantic Obstruction calculus (C1), the Residue Thesis (R1), and the companions T1 (Navier–Stokes), T2 (Riemann), T3 (Yang–Mills), and T4 (Hodge), whose tetralogy this paper extends to a pentalogy and closes.

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Tier discipline throughout: **Tier I** is standard checkable mathematics, asserted with complete proof or standard citation; **Tier II** is exact structure read after the Tier-I objects are fixed; **Tier III** is declared direction. Falsification surfaces in §9.

0. Origin

The originating intuition, at its fifth and final station:

Each of these questions is of the same form — the way the question is posed is what generates the outcome. The issue is not domain-dependent; it is structural. And P versus NP is conceptually the purest form of all the exemplars of this structural obstruction — elusive in proof for that very reason.

This paper proves the purity claim in two exact senses, each a theorem. **First (Lemma 2.1):** P vs NP is the only problem in the pentalogy in which *both* verdicts are tail predicates — neither $P = NP$ nor $P \neq NP$ admits a finite witness, even in principle; every other domain had at least one window-witnessable side (the cycle for Hodge, the counterexample for Riemann), while here even the exhibited algorithm certifies nothing without a Π -shaped closure over all inputs. **Second (Theorem 3.4, the self-reference remark made exact):** the seam this problem lives on — the coupling of local verification to global construction — is the very seam on which any *proof about the problem* must itself run: proving a separation is a construction task whose difficulty is governed by the hardness being asserted, and the domain's deepest barrier (natural proofs) is literally *powered by* that hardness.

The question and its metaquestion coincide. That coincidence is the structural reason the problem is elusive — the origin statement's "for that very reason," earned.

The paper also performs the pentalogy's standing duty to the corpus: the Λ -NSAF argument of the musings is replayed minus its error — an error the corpus's own Part 2 had already begun to name — and its sound core is retained at indexed strength (§6). Nothing below asserts $P = NP$, $P \neq NP$, or their independence; everything below is the proof of what deciding them costs.

1. Setting

1.1 The statement and the two verdicts

P : languages decidable by a deterministic Turing machine in time polynomial in input length. NP : languages with polynomial-size certificates verifiable in polynomial time. The Clay problem: decide whether $P = NP$.

Tier I (logical form). Fix the standard enumeration of machine– polynomial pairs (M, p) . "M decides SAT correctly within time p " is a Π^0_1 condition (a universal statement over inputs whose instances are time-bounded, hence decidable computations). Therefore:

- **$P = NP$ is Σ^0_2 :** $\exists(M, p) \forall x [M \text{ decides } x \text{ correctly within } p(|x|)]$.
- **$P \neq NP$ is Π^0_2 :** $\forall(M, p) \exists x [\text{failure}]$, each failure decidable.

Neither side is Σ^0_1 . This single fact drives Lemma 2.1.

1.2 The frames

Definition 1.2 (Complexity frames). A *frame* is a finite station of any of the following circumscription continua: the *algorithm-search continuum* (enumerate programs, test on inputs of bounded length); the *lower-bound continuum* (proven circuit or communication lower bounds for restricted models — bounded depth, monotone basis, algebraic degree — or for general circuits up to small linear size); the *instance continuum* (empirical hardness of solvers at accessible sizes); the *fine-grained continuum* (conditional reductions and thresholds under hypotheses such as SETH). A *window observable* is any functional of such a station. The record stations are Tier-I facts: general circuit lower bounds for explicit functions stand at small linear size; exponential lower bounds are proven in the bounded-depth and monotone frames; industrial SAT solvers dispatch enormous instances while worst-case complexity remains open.

Definition 1.3 (Frame-internal decidability; weld). As in T1–T4 verbatim.

2. Classification: both verdicts are tails — the purity lemma

Lemma 2.1 (No finite witness on either side; Tier I). Neither verdict of P vs NP is witnessed by any frame. $P \neq NP$ is Π_2^0 : its confirmation consumes the entire machine enumeration. $P = NP$ is Σ_2^0 , and — the decisive point — its existential witness is *not* a finite certificate: exhibiting an algorithm decides nothing, because the algorithm's correctness and polynomial running time are themselves Π -shaped tail conditions over all inputs. The domain even possesses a canonical explicit candidate — Levin's universal search algorithm, which is optimal for SAT-search up to constant factors, so that $P = NP$ holds *iff this one fixed program runs in polynomial time* — and the candidate's exhibition, far from closing the Σ side, converts the entire problem into a tail property of a single object. No other problem in the pentalogy has this feature: Hodge's cycle and Riemann's counterexample are finite certifiable witnesses; here the witness side is tail-infected too. **P vs NP is the pentalogy's unique doubly-blind frame: the circumscription continuum can neither confirm nor refute either verdict at any station.** This is the first exact sense of "purest form."

Proof. The quantifier computation of §1.1; Levin (1973) for the universal-search reduction; the observation that verifying "runs in time p on all inputs" is Π_1^0 , hence no finite station of testing certifies it. ■

Remark 2.2 (The window is treacherous here by theorem; Tier I). As in each companion, the domain exhibits its own proof that restricted-frame and empirical evidence does not extrapolate:

1. **The Tardos function.** Razborov proved superpolynomial monotone circuit lower bounds for the clique function — a rigorous exponential hardness verdict *in the monotone frame* — and Tardos then exhibited a function computable in polynomial time whose monotone circuit complexity is exponential. Restricted-frame hardness coexists with actual easiness: the frame's verdict does not transfer. The exact counterpart of Davenport–Heilbronn (T2) and the abelian counterfeit (T3).
2. **Bounded depth.** Parity requires exponential-size AC^0 circuits (Håstad), yet is trivially in P : depth-frame hardness transfers nothing.
3. **The solver record.** Modern SAT solvers dispatch industrial instances with millions of variables while the worst case stands open — the empirical window points both ways at once and decides neither.
4. **The hierarchy contrast.** Diagonalization *does* decide tails where it applies — $P \neq EXP$ is a theorem — and P vs NP sits exactly where Baker–Gill–Solovay proves diagonalization-type reasoning goes blind: the open problem is located precisely at the boundary of the one frame-internal weld the field possesses.

3. The obstruction theorems: the field that proved its own bars

P vs NP is unique in the pentalogy in a second way: **the domain's principal rigorous outputs are its own circumscription theorems.** The "barriers" of complexity theory are not folklore or heuristics; they are proven

theorems of exactly the B/C/D form, established by the field about itself, decades before this program named the form. This section identifies them, as T4 identified the Compiled Edition's instrument theorems.

Theorem B⁵ (Frame-blindness: the three barriers and the restricted counterfeits)

Theorem B⁵ (Tier I; = the classical barrier theorems, identified).

1. **(Relativization; Baker–Gill–Solovay 1975.)** There exist oracles A, B with $P^A = NP^A$ and $P^B \neq NP^B$. Hence no argument whose steps are invariant under relativization — the entire classical diagonalization/simulation toolkit — can decide the verdict: the frame's evidence coexists with both conclusions, by exhibited witness, on both sides simultaneously.
2. **(Natural proofs; Razborov–Rudich 1997.)** Any lower-bound proof proceeding through a property of Boolean functions that is *constructive* (efficiently decidable) and *large* (satisfied by a random function) would break exponentially hard pseudorandom generators. Conditional on standard cryptographic hardness, the entire class of natural separating properties — which includes every combinatorial lower-bound technique of the restricted-frame successes — is barred from the general frame. In the calculus's vocabulary: an anti-separation theorem for a named class of obstruction functionals, the exact analogue of the Compiled Edition's Theorem 17.6.
3. **(Algebrization; Aaronson–Wigderson 2009.)** The arithmetization technique that evaded relativization (IP = PSPACE era) is itself barred: algebrizing arguments cannot resolve P vs NP, again by two-sided oracle witnesses.
4. **(Restricted-basis counterfeits.)** Monotone and bounded-depth lower bounds do not transfer (Remark 2.2, items 1–2): the Tardos function is the standing witness that frame-internal hardness verdicts are frame-bound.

Proof. Citations as stated; the identification with the pentalogy's frame-blindness form is definitional under 1.2–1.3. ■

Theorem C⁵ (Windows buy windows)

Theorem C⁵ (Tier I/II). (i) Program testing to any input bound decides nothing (Lemma 2.1). (ii) Restricted-frame lower bounds decide their frames completely and the general frame not at all (B⁵.4). (iii) The general-circuit lower-bound ladder advances by linear-size increments — window-to-window motion, with the gap to superpolynomial untouched by any existing theorem. (iv) The analogue of the height/effectivity criterion: Levin's universal search is the domain's *delivered reformulation weld* — it reduces the entire problem to the running time of one explicit machine — and, exactly as the residue lens predicts, the reduction relocates the tail without shortening it: the one machine's polynomiality is Σ^0_2 -hard information exactly as before. A criterion converting bounded testing of that machine into the verdict would be the resolution itself.

Proof. (i)–(iii) as cited; (iv) Levin (1973) plus the quantifier count of §1.1. ■

Theorem D⁵ (The escape routes)

Theorem D⁵ (Tier I/II). (i) (*No Σ^0_1 collapse — the shortcut provably absent.*) In T2, Σ^0_1 -completeness converted non-refutability into truth; here neither verdict is Σ^0_1 (Lemma 2.1), so the D' mechanism is unavailable *in principle*: even the metamathematical shortcut that arithmetic granted Riemann is closed to P vs NP — a third purity marker. (ii) (*The forcing lane, closed unconditionally.*) P vs NP is an arithmetic statement; forcing extensions preserve arithmetic truth; its truth value is forcing-invariant, and Cohen-style independence is impossible — as in T2 Theorem D'(3), here without any coding condition. (iii) (*The four-cell lattice.*) The Compiled Edition's verdict lattice and its Con(ZFC) refutation of "unprovable $\implies$ provably unprovable" apply verbatim: the open cell is ineliminable by theorem, and any provable-independence claim must proceed by arithmetic-incompleteness methods for which no attachment to P vs NP exists or is indicated.

Proof. (i) §1.1; (ii) arithmetic invariance of forcing; (iii) Compiled Edition §24, cited. ■

Theorem 3.4 (The self-referential seam — the second purity theorem)

Theorem 3.4 (Tier II, assembled from Tier-I components). The seam of this problem — the coupling of local verification (the L-ladder) to global construction (the C-ladder) — is the seam on which every proof about the problem must itself run, in the following exact senses:

1. A proof of $P \neq NP$ is a *constructed object* whose discovery is a search problem over proof space, while its *verification* is polynomial in its length — the P-vs-NP structure instantiated at the meta level; if $P = NP$, proof search itself is easy for statements with short proofs, so the metaproblem's difficulty is entangled with the object problem's answer.
2. The natural-proofs barrier (B⁵.2) is *powered by hardness*: it bars the natural techniques exactly insofar as hard pseudorandom functions exist — the obstruction to proving hardness is a consequence of hardness. A world in which the barrier fully operates is a world in which a cousin of the conclusion is true; the wall and the treasure are made of the same material.
3. Consequently the frame and the content coincide: choosing a representation class in which to attack the problem is an instance of the very L/C coupling the problem interrogates.

This coincidence — unique in the pentalogy — is the exact structural content of the origin claim that P vs NP is the purest exemplar and "elusive in proof for that very reason": in the other four domains the circumscription obstruction constrains proofs *about* the seam; here it constrains them *because they are made of* the seam. ■ (assembly)

Corollary F⁵ (The circumscription obstruction, computational form)

Corollary F⁵ (Tier II). The P vs NP verdict is not frame-internally decidable: witnessed on neither side by any station (Lemma 2.1, the doubly-blind frame); not extrapolable from restricted frames or empirical records, by exhibited counterfeits (Remark 2.2); barred to relativizing, natural, and algebrizing techniques by the field's own theorems (B⁵); reformulable but not shortenable (C⁵); metamathematically closed on the forcing lane and shortcut-free on the Σ^0_1 lane (D⁵); and self-referential at the meta level (3.4). Every decision factors through a weld that is non-relativizing, non-natural, non-algebrizing, and frame-transferring — a conjunction of evasions that the

domain has satisfied exactly once (§4). The station at which one stops testing, bounding, or benchmarking and pronounces is, absent such a weld, arbitrary. ■

4. The weld taxonomy, instantiated

Type I — Criteria and reformulation welds

- **Cook–Levin and the Karp reductions:** the domain's transfer weld, proven and complete — thousands of problems welded into a single verdict. NP-completeness is to this problem what the explicit formula is to Riemann and Osterwalder–Schrader reconstruction to Yang–Mills: the bridge exists; the object is missing. **(W4⁵) is delivered.**
- **Levin's universal search:** the residue-lens reformulation — the verdict as the runtime of one machine (C⁵.iv).
- **Hardness-versus-randomness (Nisan–Wigderson; Impagliazzo–Wigderson):** equivalence welds coupling derandomization to circuit lower bounds — theorem-grade relocations that interlock the verdict with a second family of tails.
- **The corpus's Localization Theorem** (DST fine-grained traversal): within its declared representation class, crossing the Chen threshold for Integer Max-IP at superlogarithmic-power dimension is equivalent to resolving a single named open object — a delivered localization weld of exactly this type, relocating a conditional frontier onto one coordinate.

Type II — Invariant welds (progress measures)

- **The polynomial method (Razborov–Smolensky), rank and degree arguments, random restrictions (Håstad):** invariant-based separations — executed, rigorously, in every restricted frame where the invariant survives; barred from the general frame by natural proofs precisely because these invariants are constructive and large. The barrier does not say invariants fail; it says the *next* invariant must be of a different kind — non-constructive, or small — which is a specification, not a despair.

Type III — Carrier welds: executed, programmatic, and negative

- **The hierarchy theorems:** diagonalization as a genuine weld deciding genuine tails ($P \neq EXP$; $NP \neq NEXP$) — the proof that the fate is achievable where the technique is not barred, and the calibration of exactly where the bars begin.
- **Williams (2011): $NEXP \not\subseteq ACC^0$.** The modern exhibit and the pricing standard for (W2⁵): the *algorithmic method* welds a window object — a marginally-faster-than-brute-force circuit-satisfiability algorithm — to a tail object, a nonuniform lower bound, through a nonrelativizing, non-natural, non-algebrizing argument. Executed once, at full rigor, evading all three barriers simultaneously: the existence proof that Corollary F⁵'s conjunction of evasions is satisfiable.

- **Geometric Complexity Theory (Mulmuley–Sohoni):** the explicit carrier-change program — complexity classes as orbit closures, separations as representation-theoretic obstructions; the domain's Hilbert–Pólya/Deninger analogue, moving the permanent-versus- determinant avatar into algebraic geometry. And within it, the Bürgisser–Ikenmeyer–Panova theorem — occurrence obstructions do not suffice — stands as the executed *negative sub-door closure*, the exact counterpart of Nečas–Růžička–Šverák for Navier–Stokes profiles: a named obstruction class inside the carrier program, closed by theorem, the program continuing through the doors that remain.
- **The barriers themselves,** read as executed negative welds: Baker–Gill–Solovay, Razborov–Rudich, and Aaronson–Wigderson are rigorous proofs that whole functional classes cannot separate — the richest anti-separation corpus in mathematics, and the reason this domain supplied the Residue Thesis's cleanest Rosetta row.

5. The spine, closed over five

The table of T2–T4 §5 gains its final column. P vs NP: two coupled ladders — local verification and global construction, coupled by computation itself; verdict's logical form — Σ^0_2 against Π^0_2 , **both tails, the doubly-blind frame**; frame-internal bar — the three barrier theorems plus the restricted-frame counterfeits, the field's own circumscription theorems (B^5 – D^5); window record, verdict-silent — restricted-frame exponential bounds, linear general bounds, solver records pointing both ways; weld record — transfer delivered (NP-completeness), one full three-barrier-evading weld executed (Williams), an explicit carrier program with an internal door-closure (GCT; Bürgisser–Ikenmeyer–Panova), hierarchy welds calibrating the bars' edge.

The pentalogy's closing identity (Tier II). Five domains, one theorem-shaped structure: an exact global verdict on the seam of two coupled ladders; a native circumscription continuum blind to the verdict — confirmation-blind (Riemann), refutation-blind (Hodge), doubly tail-blind (Navier–Stokes, Yang–Mills), and doubly blind *on both verdicts* (P vs NP); domain theorems barring every frame-internal route; and decisions, wherever recorded, arriving only as welds. The ordering by distance-to-weld now completes: Hodge (carrier delivered on the majority stratum, one finite computation pending), Yang–Mills (transfer proven, carrier partially built), Riemann (transfer proven, carrier programs live), Navier–Stokes (transfer cheap, profile programs live with the Euler exhibits adjacent), and **P vs NP farthest** — both verdict-sides tail, the shortcut lanes provably absent, the barrier corpus most comprehensive, and the seam self-referential. That the purest instance is the most distant is not an accident of the ordering; by Theorem 3.4 it is the same fact twice.

The prime-origin closure, final. The Rope-and-Sand contrast reaches its limit case here: in the home domain every factorization question closes because the object space is finite; P vs NP is the question of whether *closure itself* — the passage from checkable to findable — is uniformly cheap over all finite object spaces at once. The problem is not an instance of the home domain's aggregate; it is the aggregate of all aggregates, the question the diagrams were asking about themselves.

6. The corrected moves: the Λ -NSAF argument, replayed

The corpus (the musings, Parts 1–2) is replayed minus its error — an error whose correction the corpus's own second part had begun (Tier II ledger):

1. **The Extraction Collapse lemma is retained as Tier I.** The musings' central lemma — if $P = NP$ then every NP verification relation admits polynomial certificate *construction* — is the standard search-to-decision self-reducibility argument, correct and exact. The corpus derived it cleanly.
2. **The L-frame/C-frame reading is retained as the seam.** Verification as local relation, construction as global traversal: this is precisely the two-ladder structure of §5, and identifying it was the traversal's genuine contribution — it is the reason the domain slots into the pentalogy at all.
3. **The Λ -axiom step is corrected by the control test (the E' discipline of T2, applied).** Part 1 concludes $P \neq NP$ because the $L \rightarrow C$ identification "without residue" would violate Λ , taken as an axiom. Unindexed, that axiom is *false by counterexample*: perfect matching (Edmonds), 2-SAT, linear programming, and spanning connectivity are verification relations with exponentially large candidate spaces and globally coherent certificates whose construction is nevertheless polynomial — the $L \rightarrow C$ identification closes without residue, under the same comparator, on infinitely many problems. A mechanism that fires on Edmonds's matching algorithm certifies nothing about SAT: the mandatory control experiment, exactly as the RH draft's Λ -mechanism was controlled against the Prime Number Theorem. Indexed correctly, the statement "the $L \rightarrow C$ identification for SAT leaves irreducible residue" is not an axiom from which $P \neq NP$ follows — **it is $P \neq NP$** , verbatim, in the calculus's vocabulary: whether $I_R > 0$ for the construction target over the polynomial-comparator class is the question. Assuming Λ for SAT assumes the conclusion; the argument is the purest instance in the whole corpus of the residue asserted where it is to be proven.
4. **The corpus's self-correction is credited and completed.** Part 2's status disclaimer and its translation into accepted terms — noting that the "irreducible mismatch" claim *is equivalent to* the separation claim — already performed the demotion from proof to reformulation weld. T5 completes it: the Λ -reading of P vs NP is a correct Type-I relocation of the verdict into the calculus's language (and a good one — it is the Rosetta row of R1), carrying exactly the verdict's own weight and no more.
5. **The Localization Theorem is retained at delivered strength** (§4, Type I): a genuine conditional-frontier weld within its declared class, the corpus's rigorous product in this domain.
6. **What the intuition earns, restated.** The instinct that this problem is the purest exemplar, and elusive for that very reason, is proven above (Lemma 2.1; Theorem 3.4) — the corpus's compass was correct; only the step from compass to conclusion required the weld it was pointing at.

(The two interactive explorations hosted on the GEOMETER path remain inaccessible to automated reading under the site's robots policy and are not required for the above; they can be incorporated into a revision on up-load.)

7. The priced specification: what a resolution must produce

By Corollary F⁵ and §4, a resolution must be a weld that is simultaneously non-relativizing, non-natural, and non-algebrizing, and whose conclusion is frame-transferring (general-model, not restricted-basis). The mandatory components:

(W1⁵) The carrier. A representation of computations in which the separation becomes an obstruction of a compact, certifiable kind — the GCT orbit-closure geometry, the algorithmic-method coupling of algorithms to lower bounds, or a successor evading the Bürgisser–Ikenmeyer–Panova sub-door closure. *Priced by:* GCT (the program explicit, with one interior door closed by theorem) and Williams (the coupling executed).

(W2⁵) The construction. Execution of the carrier's mechanism at the required strength: for the algorithmic method, sufficiently strong satisfiability algorithms for the general class (the delivered ACC⁰ instance is the template); for GCT, representation-theoretic obstructions beyond occurrence obstructions. *Priced by:* Williams 2011 — the one existing artifact satisfying all three evasions at once.

(W3⁵) The separation mechanism. For $P \neq NP$: an invariant — necessarily non-constructive or non-large — computing a superpolynomial lower bound for an explicit NP function in the general frame; the natural- proofs theorem is its exact negative specification. For $P = NP$: a polynomial-time SAT algorithm *with a proof* of its correctness and time bound — the Π -closure that Lemma 2.1 shows is the actual content of the Σ side; Levin's machine stands ready as the canonical object of any such proof. *Priced by:* the restricted-frame executions (Razborov, Håstad, Smolensky) as proof the invariant genre is real, with the barriers marking precisely what the general-frame instance must evade.

(W4⁵) The transfer. Cook–Levin and the completeness apparatus — **delivered**, classical, complete: any weld executed on SAT propagates to the entire class instantly.

Necessity, relative to §3: without (W1⁵) both tails stand uncompactified (Lemma 2.1); without (W3⁵)'s evasions, B⁵ revives clause by clause; (W4⁵) exists. **This is the missing weld, stated as a work order — and, for the fifth and final time, it is demonstrably the work order the field's frontier is already executing, from the algorithmic method to GCT, which is the structural reading's closing confirmation.**

8. What this paper proves, in one paragraph

That P vs NP is the pentalogy's purest instance by theorem twice over — both verdicts tail-shaped, the frame doubly blind, even the exhibited Levin witness converting the problem into a tail property of one object (Lemma 2.1); and the seam self-referential, proofs about the problem being made of the problem's own material, with the deepest barrier powered by the hardness it obstructs (Theorem 3.4); that the frame-internal bars are the field's own celebrated theorems, identified — relativization, natural proofs, algebrization, and the restricted-frame counterfeits led by the Tardos function (Theorem B⁵, Remark 2.2); that windows buy windows, with Levin search as the

delivered reformulation that relocates without shortening (Theorem C⁵); that the escape lanes are closed or absent — forcing barred unconditionally by arithmetic invariance, the Σ^0_1 shortcut of the Riemann case provably unavailable (Theorem D⁵); that the weld taxonomy is instantiated with the domain's distinctive completeness — transfer delivered by Cook–Levin, one full three-barrier-evading weld executed by Williams, a carrier program explicit in GCT with an interior door closed by Bürgisser–Ikenmeyer–Panova, and the barrier corpus itself standing as the richest set of executed negative welds in mathematics (§4); that the corpus's Λ –NSAF argument, controlled against Edmonds's matching and indexed by the calculus, is corrected from proof to reformulation — its lemma sound, its seam-identification retained, its axiom recognized as the theorem it sought (§6); and that a resolution must satisfy the specification (W1⁵)–(W3⁵) over the delivered transfer (W4⁵), completing the pentalogy's spine with P vs NP at the greatest distance-to-weld — the purest instance and the farthest, by the same fact stated twice (§5, §7).

9. Standing and falsification surfaces

Asserted as proven: Lemma 2.1; Theorems 2.2 (via its cited exhibits), B⁵, C⁵, D⁵, 3.4 (Tier II assembly of Tier-I parts), Corollary F⁵; the taxonomy classifications and the spine closure (Tier II over Tier-I rows); the necessity direction of §7.

Not asserted: P = NP, P ≠ NP, their independence, or the achievability of (W1⁵)–(W3⁵); any claim that the corpus's *intuition* was unsound — §6 and Theorem 3.4 argue the contrary at theorem grade.

To defeat this paper, exhibit one of:

- **F1⁵** — an error in the proof or identification of B⁵, C⁵, D⁵, or 3.4;
- **F2⁵** — a finite witness for either verdict (refutes Lemma 2.1), or a relativizing/natural/algebrizing resolution (refutes B⁵ by refuting a barrier theorem);
- **F3⁵** — a bounded-testing criterion for Levin's machine not equivalent to the resolution (refutes C⁵.iv);
- **F4⁵** — a rigorous separation or collapse verdict in complexity theory outside the three-type taxonomy;
- **F5⁵** — a valid resolution lacking (W1⁵) or (W3⁵), or an unconditional structural-hardness mechanism that provably does not fire on Edmonds's matching, 2-SAT, and linear programming, together with its validity (engages §6.3 at the control surface, as any serious successor to the Λ -argument must).

Absent such an exhibit, the origin statement stands proven at its fifth station and over the completed pentalogy: *the way the question is posed is what generates the outcome; P versus NP is the purest form of the structural obstruction — both of its verdicts tail, its frame doubly blind, its seam self-referential — and elusive in proof for exactly that reason; the issue is not domain-dependent; it is structural; and in all five domains, every road to a verdict runs through a weld whose specification is now complete, whose executed exemplars are named, and whose historical record contains no exception.*

References and source anchors

Program corpus. Semita, L., the Λ -NSAF P vs NP musings, Parts 1–2 (the object of §6); the DST fine-grained traversal and Localization Theorem; *The Hodge Traversal, Compiled Edition V4–V8*; *The Obstruction Calculus, Consolidated (C1)*; *The Residue Thesis (R1)*; T1–T4 of this series; *The Rope-and-Sand Gambit* (PRIMES repository).

Foundations and form. Cook (1971); Levin (1973), universal search and NP-completeness; Karp (1972); the arithmetical-hierarchy placement of P vs NP (folklore; see Aaronson's survey $P \stackrel{?}{=} NP$).

Barriers (Theorem B⁵). Baker–Gill–Solovay (1975); Razborov–Rudich (1997), natural proofs; Aaronson–Wigderson (2009), algebrization.

Restricted-frame welds and counterfeits. Razborov (1985), monotone clique; Tardos (1988), the monotone/general gap; Håstad (1986), AC^0 lower bounds for parity; Razborov–Smolensky, the polynomial method.

Executed and programmatic carriers. Time hierarchy (Hartmanis–Stearns); Williams (2011), $NEXP \not\subseteq ACC^0$ and the algorithmic method; Mulmuley–Sohoni, Geometric Complexity Theory; Bürgisser–Ikenmeyer–Panova (2016/2019), no occurrence obstructions; Nisan–Wigderson; Impagliazzo–Wigderson, hardness versus randomness.

Control witnesses (§6.3). Edmonds (1965), maximum matching in polynomial time; 2-SAT (Krom; Aspvall–Plass–Tarjan); Khachiyan (1979), linear programming.

Logic. Arithmetic invariance under forcing; Gödel; the Compiled Edition's verdict lattice (§24) with the Con(ZFC) witness.